



विशाल जाधव सरांचे

**VJTech Academy**

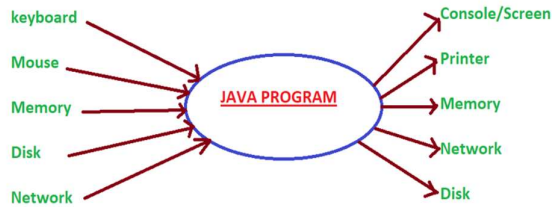
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## “File Handling in JAVA”

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**Live Class Lectures**

## Managing Input/Output Files in JAVA



- 1) System.in (Input Stream)
- 2) System.out (Output Stream)
- 3) System.err (Error Stream)

java.io package

### Streams:

- Collection of continues group of data which will travel from one point to another point is known as Stream.



(a) Reading data into a program.



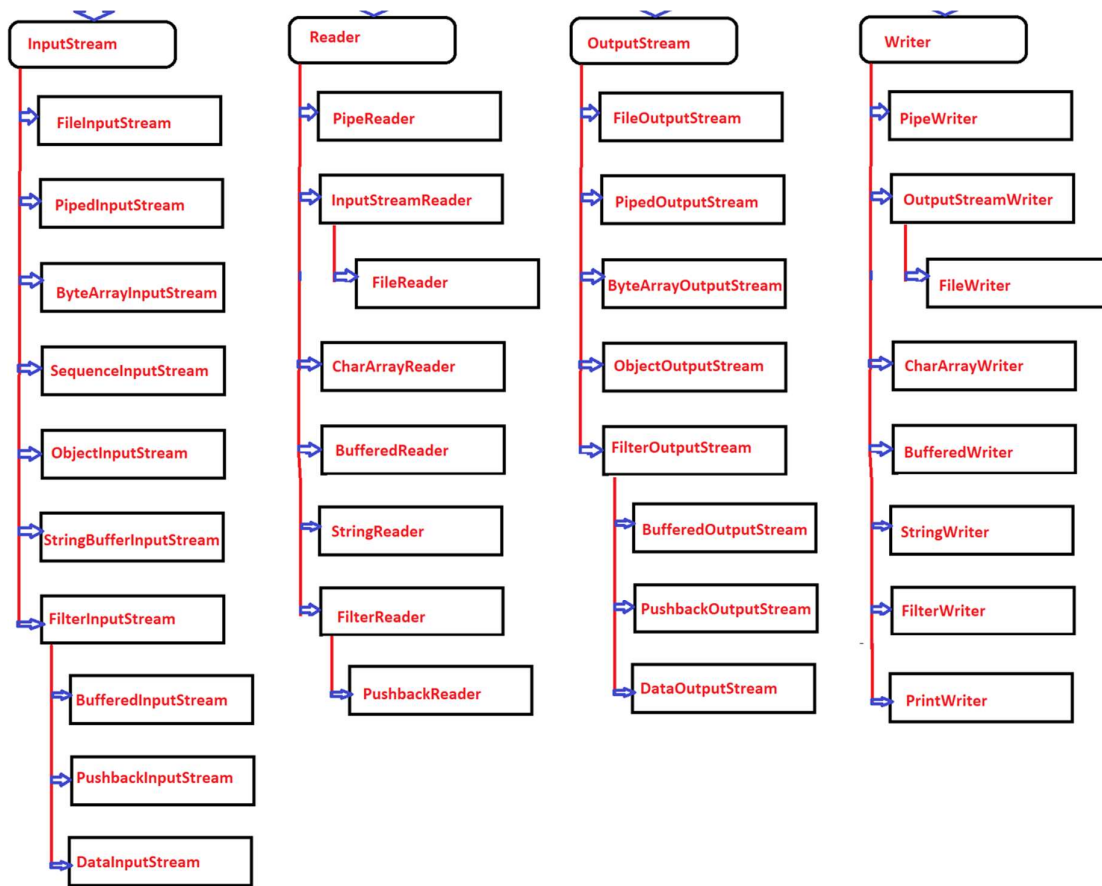
(a) Writing data to destination

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| Byte Stream Class                                                              | Character Stream Class                                                        |
|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| 1) Provides support for handling IO operations on <u>byte</u> .                | 1) Provides support for handling IO operations on <u>characters</u> .         |
| 2) Two types:<br>i - Input Stream Classes<br>ii - Output Stream Classes        | 2) Two types:<br>i - Reader Classes<br>ii - Writer Classes                    |
| 3) Input stream classes are used to read 8 bit byte data.                      | 3) Reader stream classes are used to read character from files.               |
| 4) Output stream classes are used to write 8 bit byte data.                    | 4) Writer stream classes are used to write character into files.              |
| 5) We can translate character stream into byte stream with OutputStreamWriter. | 5) We can translate byte stream into character stream with InputStreamReader. |

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| InputStream Class                                                                                                                                                                                                                                                                                                                                                                                                            | Reader Class                                                                                                                                                                                                                                                                                                                                                                                                | OutputStream Class                                                                                                                                                                                                                                                                                                                        | Writer Class                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1) InputStream classes are used to read 8 bit bytes data.                                                                                                                                                                                                                                                                                                                                                                    | 1) Reader class is used to read 16 bit unicode character stream.                                                                                                                                                                                                                                                                                                                                            | 1) OutputStream classes are used to write 8 bit bytes data.                                                                                                                                                                                                                                                                               | 1) Writer class is used to 16 bit unicode character stream.                                                                                                                                                                                                                                                                           |
| 2) InputStream classes are byte-oriented.                                                                                                                                                                                                                                                                                                                                                                                    | 2) Reader class is character oriented.                                                                                                                                                                                                                                                                                                                                                                      | 2) OutputStream classes are byte-oriented.                                                                                                                                                                                                                                                                                                | 2) Writer class is character oriented.                                                                                                                                                                                                                                                                                                |
| 3) Constructor: InputStream()                                                                                                                                                                                                                                                                                                                                                                                                | 3) Constructor: Reader()<br>Reader(Object lock)                                                                                                                                                                                                                                                                                                                                                             | 3) Constructor: OutputStream()                                                                                                                                                                                                                                                                                                            | 3) Constructor: Writer()<br>Writer(Object lock)                                                                                                                                                                                                                                                                                       |
| 4) Methods:<br>i) read() - Reads the next byte of data<br>ii) read(byte b[]) - Read some bytes from input stream & stored them into byte array b.<br>iii) read(byte b[],int m,int n) - Read n bytes data from input stream starting from mth byte & stored them into byte array b.<br>iv) close() - Closes the input stream.<br>v) skip(n) - skips over n bytes from input stream<br>vi) available() - No of bytes available | 4) Methods:<br>i) read() - Read single character<br>ii) read(char c[]) - Read characters into an array.<br>iii) read(char c[],int m,int n) - Read n characters data from input stream starting from mth character & stored them into char array c.<br>iv) close() - Closes the input stream.<br>v) skip(n) - skips over n characters from stream<br>vi) reset() - Goes back to the beginning of the stream. | 4) Methods:<br>i) write() - Writes byte of data<br>ii) write(byte b[]) - Writes byte array b to the output stream.<br>iii) write(byte b[],int m,int n) - Write n bytes data from array b to output stream which is starting from mth byte<br>iv) close() - Closes the output stream<br>v) skip(n) - skips over n bytes from output stream | 4) Methods:<br>i) write(char c) - write character c<br>ii) write(char c[]) - writes character array.<br>iii) write(char c[],int m,int n) - write n characters from array c starting from m character.<br>iv) write(String str) - writes string.<br>v) close() - closes the writer stream.<br>vi) flush() - flushes the output stream. |



### file read operation using FileReader class

```
import java.io.*;
class FileReaderDemo
{
    public static void main(String args[])throws IOException
    {
        FileReader fin=new FileReader("vjtech.txt");
        int x;
        while((x=fin.read())!=-1)
        {
            System.out.print((char)x);
        }
        fin.close();
    }
}
```

### file write operation using FileWriter class

```
import java.io.*;
class FileWriterDemo
{
    public static void main(String args[])throws IOException
    {
        FileWriter fout=new FileWriter("msbte.txt");
        String str="Welcome to world of java programming";
        fout.write(str);
        System.out.println("File Written Successfully");
        fout.close();
    }
}
```

### count no of white spaces in file

```
import java.io.*;
class CountNoOfWhiteSpacesInFile
{
    public static void main(String args[])throws IOException
    {
        FileReader fin=new FileReader("vjtech.txt");
        int x,count=0;
        while((x=fin.read())!=-1)
        {
            char y=(char)x;
            if(y==' ')
            {
                count++;
            }
        }
        System.out.println("No of white spaces present in file="+count);
        fin.close();
    }
}
```

### count no of alphabet 'a' in file

```
import java.io.*;
class CountNoOfAlphabetA
{
    public static void main(String args[])throws IOException
    {
        FileReader fin=new FileReader("vjtech.txt");
        int x,count=0;
        while((x=fin.read())!=-1)
        {
            char y=(char)x;
            if(y=='a')
            {
                count++;
            }
        }
        System.out.println("No of alphabet a present in file="+count);
        fin.close();
    }
}
```

### count no of characters in file

```
import java.io.*;
class CountNoOfCharacters
{
    public static void main(String args[])throws IOException
    {
        FileReader fin=new FileReader("vjtech.txt");
        int x,count=0;
        while((x=fin.read())!=-1)
        {
            count++;
        }
        System.out.println("No of characters present in file="+count);
        fin.close();
    }
}
```



### Copy contents of one file to another file

```
import java.io.*;
class CopyContentsDemo
{
    public static void main(String args[])throws IOException
    {
        FileReader fin=new FileReader("vjtech.txt");
        FileWriter fout=new FileWriter("pune.txt");
        int x;
        while((x=fin.read())!=-1)
        {
            fout.write((char)x);
        }
        System.out.println("File Contents Copied Successfully");
        fin.close();
        fout.close();
    }
}
```

### file read operation using FileInputStream class

```
import java.io.*;
class FileInputStreamDemo
{
    public static void main(String args[])throws IOException
    {
        FileInputStream fin=new FileInputStream("vjtech.txt");
        int x;
        while((x=fin.read())!=-1)
        {
            System.out.print((char)x);
        }
        fin.close();
    }
}
```

### file write operation using `FileOutputStream` class

```
import java.io.*;
class FileOutputStreamDemo
{
    public static void main(String args[])throws IOException
    {
        FileOutputStream fout=new FileOutputStream("msbte.txt");
        byte b[]="VJTech Academy".getBytes();
        fout.write(b);
        System.out.println("File Written Successfully");
        fout.close();
    }
}
```

### file basic operations

```
import java.io.*;
class FileDemo
{
    public static void main(String args[])throws IOException
    {
        File f1=new File("F:/JavaBatch2024/File Handling","vjtech.txt");
        if(f1.exists())
        {
            System.out.println("File is present");
            System.out.println("File Name:"+f1.getName());
            System.out.println("Parent Name:"+f1.getParent());
            System.out.println("Length of file:"+f1.length());
            System.out.println("is file:"+f1.isFile());
            System.out.println("is Directory:"+f1.isDirectory());
            System.out.println("can read:"+f1.canRead());
            System.out.println("can write:"+f1.canWrite());
            System.out.println("Get path:"+f1.getPath());
        }
        else
        {
            System.out.println("File is not present");
            f1.createNewFile();
            System.out.println("File created successfully");
            File f2=new File("F:/JavaBatch2024/Corona");
            f2.mkdir();
            System.out.println("Directory created successfully");
        }
    }
}
```

### OUTPUT

```
File is present
File Name:vjtech.txt
Parent Name:F:\JavaBatch2024\File Handling
Length of file:6901
is file:true
is Directory:false
can read:true
can write:true
Get path:F:\JavaBatch2024\File Handling\vjtech.txt
F:\JavaBatch2024\File Handling>java FileDemo
File is not present
File created successfully
Directory created successfully
```